OPTICS · SEMESTER 5 · MODULE 2

Prepare: Interference

Everything you need before you open the question paper, from Plus Two up to Ghatak §13.5, 14.1, 14.3–14.12, 15.1–15.4, 15.8–15.11.

File 1 of 3	**Prepare** (this file): learn the ideas
File 2	Model paper: PYQs + new questions
File 3	Answer key: full solutions

Each answer in File 3 starts with topic codes such as T12 P4 M1. If a code is new to you, read that topic here first.

How to read this file

One topic at a time, in order. Each topic is small; do not skip the boxes.

REMIND ME

Facts from Class 10, Plus One or Plus Two that you have probably forgotten.

MATHS TOOL

A small piece of maths, explained the first time it is needed.

THE IDEA

The physics in plain words, before any symbols.

RESULT

What you must remember and write in the exam.

EXAM TRAP

A common mistake, and how to avoid it.

CHECK YOURSELF

Questions with the answer just below. Cover it with your thumb first.

One rule for the whole module

COUNT THE EXTRA PATH, THEN COMPARE WITH λ

Every question in this module is the same three steps. (1) Two copies of the **same** light reach a point by two routes. (2) Find how much longer one route is (the **path difference** Δ), adding half a wave for any "bad" reflection. (3) If Δ is a whole number of wavelengths the copies add (bright); if it is a whole number plus a half they cancel (dark).

Codes and map

P1–P6 Plus One / Plus Two bridge

M1–M4 Maths tools

T1–T21 Module 2 topics

Q12 question 12 of the model paper

SYLLABUS UNIT	TOPICS
5. Superposition of two sinusoidal waves	T1
6. Division of wavefront: introduction	T2
7. Interference of light waves	T3, T4, T5, T6
8. Fresnel's two mirrors, biprism	T7, T8
9. White light, Lloyd's mirror, phase change on reflection	T9, T10, T11
10. Division of amplitude, non-reflecting films	T12, T13, T14, T15
11. Colours of thin films, Newton's rings, Michelson	T16–T21

25 of the 70 external marks: the biggest module. Newton's rings, Michelson, thin films and the two-beam intensity formula are asked almost every year.

The whole module in one story

THE IDEA

1. Two waves at one place simply add (superposition). In step: bright. Opposite: dark (T1).

2. Two ordinary lamps never show this, because their phases jump at random. You must split **one** source into two copies (coherent sources) (T2).

3. Splitting the wavefront (side by side): Young's slits, Fresnel's two mirrors, the biprism, Lloyd's mirror. Bright and dark bands of width λD/d (T3–T10).

4. A reflection from a denser medium adds half a wave (T11).

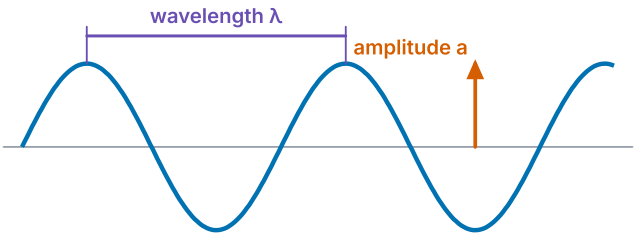
5. Splitting the amplitude (part reflected, part passed): thin films, soap bubbles, coatings on lenses, wedges, Newton's rings, Michelson's interferometer. The extra path is 2nd cos r (T12–T21).

REMIND ME · SYMBOLS USED EVERYWHERE

- **λ** wavelength (in air or vacuum)
- **Δ** path difference
- **δ** phase difference $= \frac{2\pi}{\lambda}\Delta$
- **β** fringe width (bright to next bright)
- **n** refractive index
- **m** order: 0, 1, 2, ...

P1 Waves: phase and path

> Plus One Ch 14. The language of the whole module.

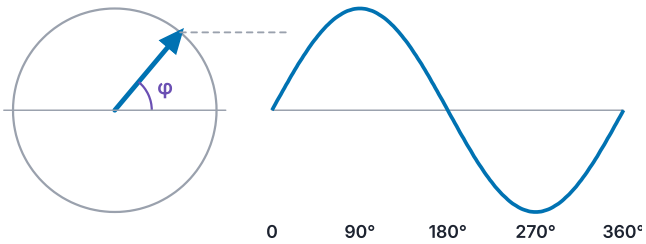


one full wave = phase 2π (360°); half a wave = π

A wave. Moving one wavelength along it changes the **phase** by 2π. So a path difference Δ means a phase difference $\delta = \frac{2\pi}{\lambda}\Delta$.

REMIND ME · PLUS ONE CH 14

- A wave: $y = a\cos(kx - \omega t)$ with $k = \frac{2\pi}{\lambda}$, $\omega = 2\pi\nu$. a is the **amplitude**, the bracket is the **phase**.
- Light has λ ≈ 4000 Å (violet) to 7000 Å (red). 1 Å $= 10^{-10}$ m $= 10^{-8}$ cm.



one full turn = one wavelength of path

Phase is an angle. A point going round the circle gives the wave (its height). One full turn (360° = 2π) is one wavelength. A path difference of λ/4 is a phase difference of 90°.

RESULT · PATH → PHASE

$$\delta = \frac{2\pi}{\lambda}\,\Delta$$

Δ = 0, λ, 2λ, ... δ = 0, 2π, 4π: in step

Δ = λ/2, 3λ/2, ... δ = π, 3π: opposite

EXAM TRAP

Do not mix up path difference (a length, in cm or Å) with phase difference (an angle, in radians). Always convert with $2\pi/\lambda$.

CHECK YOURSELF

Path difference λ/4. Phase difference?

$\frac{2\pi}{\lambda} \cdot \frac{\lambda}{4} = \frac{\pi}{2}$, i.e. 90°.

P2 Intensity = amplitude squared, averaged

Plus Two Ch 10. What the eye actually sees.

REMIND ME · PLUS TWO §10.4

- The brightness (intensity) of light is proportional to the **square** of the amplitude: $I \propto a^2$. Double the amplitude: four times as bright.
- Light vibrates about 10^{15} times a second. The eye (and any detector) responds in about 0.1 s, so it sees only the **average** over millions of vibrations.

MATHS TOOL · AVERAGES OVER TIME

Averaged over a full cycle (or many cycles):

$\langle \cos \omega t \rangle = 0$ (as much positive as negative)

$\langle \cos^2 \omega t \rangle = \frac{1}{2}$ (because $\cos^2 x = \frac{1 + \cos 2x}{2}$ and $\langle \cos 2x \rangle = 0$)

A quantity that changes at random (not a cycle) also averages to zero: $\langle \cos \phi_{\text{random}} \rangle = 0$.

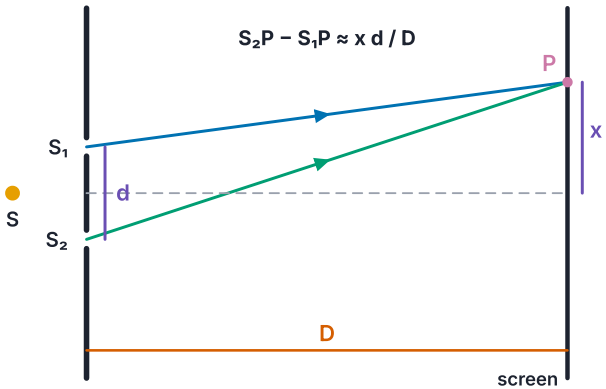
CHECK YOURSELF

Two waves give amplitude 3 units and 1 unit at two places. Ratio of intensities?

$3^2 : 1^2 = 9 : 1$.

P3 Young's experiment (as in Plus Two)

Plus Two §10.5. This module starts here and goes deeper.



Young's double-slit arrangement. Light from one slit S falls on two slits S_1, S_2 (distance d apart). They act as two coherent sources. On a screen at distance D, the point P at height x gets two waves with path difference $\Delta \approx xd/D$.

REMIND ME · PLUS TWO §10.5

- One narrow slit S lights two slits S_1, S_2 a distance d apart. A screen is at distance D ($D \gg d$).
- Bright fringe where $\Delta = n\lambda$, dark where $\Delta = (n + \frac{1}{2})\lambda$.
- Bright fringes at $y_n = \frac{n\lambda D}{d}$; fringe width $\beta = \frac{\lambda D}{d}$.
- $I = 4I_0 \cos^2(\phi/2)$; two independent sources give just $2I_0$ everywhere.

WHAT IS NEW IN THIS MODULE

Why it works (coherence), the proof of $I = 4I_0 \cos^2(\delta/2)$ for unequal waves too, and the other ways of making two coherent sources.

P4 Refraction and optical path

Class 10, Plus Two Ch 9. Needed for every thin-film question.

REMIND ME · CLASS 10 AND PLUS TWO

- Refractive index $n = \frac{c}{v}$: light is n times slower in the medium.
- Snell's law (Class 10 form): $n = \frac{\sin i}{\sin r}$ for light from air into the medium. General form: $n_1 \sin\theta_1 = n_2 \sin\theta_2$.
- The frequency stays the same, so the wavelength inside is shorter: $\lambda_{\text{medium}} = \frac{\lambda}{n}$.

THE IDEA · OPTICAL PATH

A length t inside glass contains n times as many waves as the same length of air. So for counting waves, glass of thickness t "counts as" air of length nt. This nt is the **optical path**. Path differences in this module are always optical paths.

RESULT

Optical path	$n\times$ (geometrical length)
Plate in one beam	adds $nt - t = (n-1)t$
Down and back in a film	$2nt$ (normal incidence)

CHECK YOURSELF

Glass plate, $n = 1.5$, $t = 0.01$ mm, put in one beam. Extra path?

$(1.5 - 1) \times 0.01 = 0.005$ mm $= 5000$ nm, about 10 wavelengths of green light.

P5 Mirrors, prisms and lenses

Plus Two Ch 9. Each method of making two sources uses one of these.

REMIND ME · PLUS TWO CH 9

- **Plane mirror**: the image is as far behind the mirror as the object is in front. Turning a mirror by an angle θ turns the reflected ray by 2θ.
- **Thin prism** (small angle A): every ray is bent by the same small angle $\delta = (n-1)A$.
- **Lens** (Cartesian sign convention, as in Module 1): $\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$; magnification $= v/u$.
- Reflection from a denser medium (for example air → glass) turns the wave upside down: a phase change of π. (Proved in T11.)

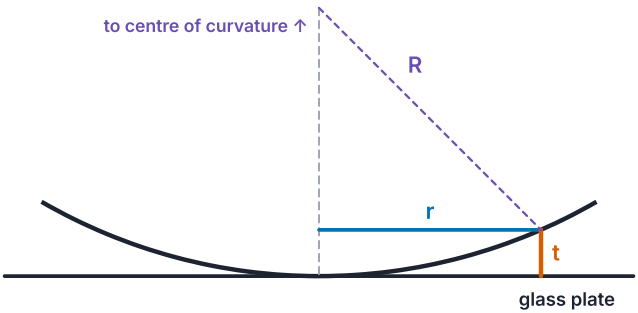
CHECK YOURSELF

A thin prism of angle 1° and $n = 1.5$. Deviation?

$(1.5 - 1) \times 1^\circ = 0.5^\circ$.

P6 Geometry of a circle

Class 10 Pythagoras, used for Newton's rings.



$R^2 = r^2 + (R - t)^2 \Rightarrow r^2 = t(2R - t) \approx 2Rt$

Newton's rings geometry. At distance r from the point of contact the air film has thickness t. Pythagoras in the big right triangle (sides r, $R - t$, hypotenuse R) gives $r^2 \approx 2Rt$.

1. A circle of radius R touches a flat line. At distance r sideways, the gap t between circle and line satisfies $R^2 = r^2 + (R - t)^2$
   Pythagoras in the right triangle: hypotenuse R, sides r and R − t.
2. Expand: $R^2 = r^2 + R^2 - 2Rt + t^2 \;\Rightarrow\; r^2 = 2Rt - t^2$
3. t is tiny compared with $2R$ (micrometres against metres), so drop t^2: $r^2 \approx 2Rt$

CHECK YOURSELF

R = 100 cm, r = 0.1 cm. Gap t?

$t = r^2/2R = 0.01/200 = 5 \times 10^{-5}$ cm, about one wavelength.

M1 Four trig identities

Plus One Maths Ch 3. These are the only identities the module uses.

MATHS TOOL

(a) $\cos(A+B) = \cos A \cos B - \sin A \sin B$

(b) $\cos A + \cos B = 2\cos\dfrac{A+B}{2}\cos\dfrac{A-B}{2}$

(c) $2\cos A \cos B = \cos(A-B) + \cos(A+B)$

(d) $1+\cos\delta = 2\cos^2\dfrac{\delta}{2}$, $1-\cos\delta = 2\sin^2\dfrac{\delta}{2}$

WHERE EACH ONE IS USED

(a) splits $\cos(\omega t+\theta)$ into a $\cos\omega t$ part and a $\sin\omega t$ part (T1).

(b) adds two equal waves in one line (T4).

(c) turns a product of two waves into a sum, so we can average it (T4).

(d) turns $2I_0(1+\cos\delta)$ into $4I_0\cos^2(\delta/2)$ (T4).

CHECK YOURSELF

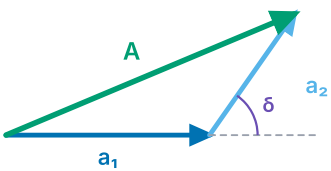
Use (b): $\cos\omega t + \cos(\omega t+\pi/2) = ?$

$2\cos\dfrac{\pi}{4}\cos(\omega t+\dfrac{\pi}{4}) = \sqrt{2}\cos(\omega t+\dfrac{\pi}{4})$: amplitude √2, not 2.

M2 Waves as arrows (phasors)

A picture that gives the T1 result without algebra.

$$A^2 = a_1^2 + a_2^2 + 2a_1a_2 \cos \delta$$



Phasor method: draw each wave as an arrow (length = amplitude, angle = phase). Join them head to tail. The arrow from start to end is the resultant. The cosine rule gives its length.

THE IDEA

Draw each wave as an arrow: length = amplitude, angle = phase. Two waves of the same frequency add like arrows placed head to tail. The resultant arrow gives the amplitude (its length) and the phase (its angle) of the sum.

1. Arrows a_1 and a_2 with angle δ between their directions. The triangle has the outside angle δ, so the inside angle is $\pi - \delta$

   Angles on a straight line add to π.

2. Cosine rule: $A^2 = a_1^2 + a_2^2 - 2a_1a_2\cos(\pi - \delta) = a_1^2 + a_2^2 + 2a_1a_2\cos\delta$

   $\cos(\pi - \delta) = -\cos\delta$.

CHECK YOURSELF

$a_1 = a_2 = a$, δ = 90°. Resultant?

$A^2 = a^2 + a^2 + 0 = 2a^2$, so $A = \sqrt{2}\,a$ (same as M1 check).

M3 Small quantities: three approximations

> Every "≈" in this module is one of these.

MATHS TOOL

Small angle	θ in **radians**, small: $\sin\theta \approx \tan\theta \approx \theta$, $\cos\theta \approx 1$
Drop the tiny	If $t \ll R$: $2Rt - t^2 \approx 2Rt$
Sum ≈ double	If $D \gg d$: $S_1P + S_2P \approx 2D$

REMIND ME · RADIANS

π rad = 180°, so 1° $= \pi/180 = 0.01745$ rad. 1 minute of arc (1′) = 1/60 degree $= 2.91 \times 10^{-4}$ rad.

CHECK YOURSELF

Is $\sin 1^\circ \approx 0.01745$?

Yes: $\sin 1^\circ = 0.017452$. The error is less than 0.01%.

M4 Units and powers of ten

> Most wrong answers in numericals are unit mistakes.

1 Å	10^{-10} m $= 10^{-8}$ cm
1 nm	10^{-9} m $= 10$ Å
1 µm	10^{-6} m $= 10^{-4}$ cm
5000 Å	$= 500$ nm $= 5 \times 10^{-7}$ m $= 5 \times 10^{-5}$ cm
589 nm	$= 5890$ Å $= 5.89 \times 10^{-7}$ m

HOW TO DO EVERY NUMERICAL

1. Write the formula in symbols. **2.** Convert every length to the same unit (all m, or all cm). **3.** Put in numbers. **4.** Ask: is the size sensible? (Fringe widths: fractions of a mm. Film thicknesses: about 10^{-5} cm.)

CHECK YOURSELF

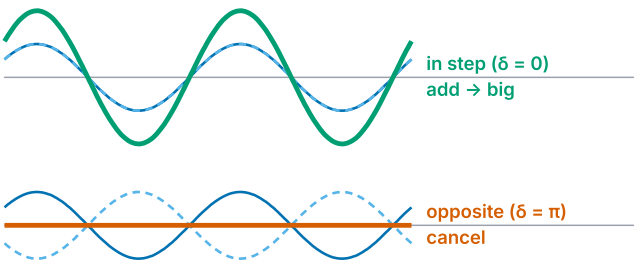
$\lambda = 6 \times 10^{-5}$ **cm,** $D = 100$ **cm,** $d = 0.1$ **cm. β?**

$6 \times 10^{-5} \times 100/0.1 = 0.06$ cm $= 0.6$ mm. Sensible.

Adding two waves

T1 Superposition of two sinusoidal waves

The principle, and the conditions for bright and dark (PYQ N19 Q15, N22 Q19, N24 Q4, N25 Q2).



blue + light blue = thick line (the sum)

Amplitudes add with their signs, point by point

Superposition: at every point the two disturbances simply add. In step they reinforce (bright). Half a wave out of step they cancel (dark).

RESULT · PRINCIPLE OF SUPERPOSITION

When two (or more) waves pass through the same point at the same time, the resultant **displacement** is the sum of the separate displacements: $y = y_1 + y_2$. Each wave carries on afterwards as if the other had never been there.

THE IDEA · INTERFERENCE

Because displacements add with their signs, two waves can make **more** than either (crest on crest) or **nothing** (crest on trough). This rearrangement of light into bright and dark regions is **interference**.

Two waves of the same frequency

$x_1 = a_1 \cos(\omega t + \theta_1)$ and $x_2 = a_2 \cos(\omega t + \theta_2)$. Find the resultant $x = x_1 + x_2$.

1. Open each bracket: $x = (a_1 \cos\theta_1 + a_2 \cos\theta_2)\cos\omega t - (a_1 \sin\theta_1 + a_2 \sin\theta_2)\sin\omega t$
   Identity (a) of M1 on each wave.

2. Try $x = a\cos(\omega t + \theta) = a\cos\theta\cos\omega t - a\sin\theta\sin\omega t$
   We guess the answer is again one wave, with amplitude a and phase θ.

3. Match the $\cos\omega t$ and $\sin\omega t$ parts: $a\cos\theta = a_1\cos\theta_1 + a_2\cos\theta_2$, $a\sin\theta = a_1\sin\theta_1 + a_2\sin\theta_2$
   True at every time t only if both coefficients agree (Ghatak 13.23–13.24).

4. Square and add: $a^2 = a_1^2 + a_2^2 + 2a_1a_2\cos(\theta_1 - \theta_2)$
   $\cos^2 + \sin^2 = 1$, and the cross terms give $\cos\theta_1\cos\theta_2 + \sin\theta_1\sin\theta_2 = \cos(\theta_1 - \theta_2)$.

5. Divide: $\tan\theta = \dfrac{a_1\sin\theta_1 + a_2\sin\theta_2}{a_1\cos\theta_1 + a_2\cos\theta_2}$

Same answer as the arrow picture (M2), with δ = $\theta_1 - \theta_2$.

RESULT · CONSTRUCTIVE AND DESTRUCTIVE

$$a^2 = a_1^2 + a_2^2 + 2a_1a_2\cos(\theta_1 - \theta_2)$$

Constructive $\theta_1 - \theta_2 = 0, 2\pi, 4\pi, \ldots \Rightarrow a = a_1 + a_2$ (largest)

Destructive $\theta_1 - \theta_2 = \pi, 3\pi, \ldots \Rightarrow a = |a_1 - a_2|$ (smallest; zero if $a_1 = a_2$)

In terms of path: constructive if $\Delta = m\lambda$, destructive if $\Delta = (m + \frac{1}{2})\lambda$.

ENERGY IS NOT LOST

Where the waves cancel, the energy is not destroyed: it appears at the bright places. Interference only **redistributes** energy. (Proof in T4: the average intensity stays $I_1 + I_2$.)

MORE THAN TWO WAVES

For many waves, add all the arrows: $a\cos\theta = \sum a_i\cos\theta_i$, $a\sin\theta = \sum a_i\sin\theta_i$ (Ghatak 13.31–13.34).

EXAM TRAP · "DO INTENSITIES OBEY SUPERPOSITION?"

No. Displacements (amplitudes) add. Intensities are squares, and $(a_1 + a_2)^2 \neq a_1^2 + a_2^2$. Equal waves in step give **4** times one intensity, not 2. Intensities add only for incoherent waves (T2, T4).

CHECK YOURSELF

Amplitudes 3 and 4, phase difference 90°. Resultant amplitude?

$a^2 = 9 + 16 + 0 = 25$, $a = 5$.

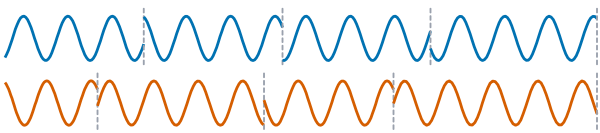
In the model paper: **Q1** **Q2** **Q25** .

Getting light to interfere

T2 Coherence

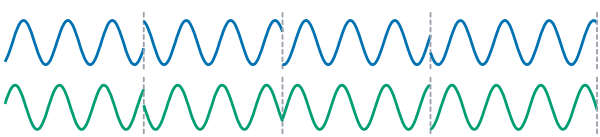
> Why two lamps never give fringes (PYQ N24 Q5, N22 Q13, N20 Q13, N25 Q3, N19 Q17).

Two separate lamps: phase jumps at random, at different times



→ phase difference changes ~10^{10} times a second: no steady fringes

One source split in two (Young, biprism, film): copies jump together



→ phase difference fixed (set only by the path difference): coherent

Light from an atom comes in short wave trains (about 10^{-10} s, about 3 cm long). Between trains the phase jumps at random. Splitting **one** train into two copies keeps their phase difference fixed.

RESULT · COHERENT SOURCES

Two sources are **coherent** if the phase difference between their waves stays **constant** in time. If the phase difference changes at random, they are **incoherent**.

Why two independent lamps give no fringes

1. An atom radiates for about 10^{-10} s (a wave train of about 10^{6} waves), then another atom takes over with a new, random phase
   Ghatak §14.4.
2. So the phase difference φ between two lamps jumps about 10^{10} times a second. The fringe pattern moves just as often
   Bright and dark swap places each time φ jumps.
3. The eye averages over 0.1 s. In $I = I_1 + I_2 + 2\sqrt{I_1 I_2}\cos\phi$, the average of $\cos\phi$ is zero
   P2: a random angle averages cos to 0.
4. Result: $I = I_1 + I_2$, uniform. Two equal lamps give $2I$ everywhere, with no fringes

THE CURE

Take **one** source and split its light into two parts. Every random jump happens in both parts at once, so their phase difference stays fixed. There are two ways to split:

Division of wavefront	Use two different **parts** of the same wavefront, side by side (Young's slits, Fresnel's mirrors, biprism, Lloyd's mirror).
Division of amplitude	Split the **same** part of the wave into two weaker beams by partial reflection (thin films, Newton's rings, Michelson).

Coherence length

THE IDEA

The two copies of one wave train must overlap when they meet. If one copy travels more than the train's length further, it meets a **different** train, and the fringes vanish. So the path difference must be smaller than the length of a wave train.

RESULT

$$L = c\,\tau_c \approx \frac{\lambda^2}{\Delta\lambda}$$

Ordinary lamp	$\tau_c \sim 10^{-10}$ s $\Rightarrow L \sim 3$ cm
Sodium light	about 1 cm
White light	a few wavelengths only
Laser	metres or more

$\Delta\lambda$ = spread of wavelengths in the light. Purer colour $\Rightarrow$ longer trains.

This is **temporal** coherence. **Spatial** coherence: the source must also be narrow, or the fringes from its different points wash each other out (T5).

RESULT · CONDITIONS FOR SUSTAINED (STEADY) INTERFERENCE

1. The two sources must be **coherent**: derived from one source.
2. They must give the **same frequency** (same wavelength).
3. Their amplitudes should be **equal or nearly equal**, so the dark fringes are really dark (good contrast).
4. They should be **narrow** (fine slits), so each fringe is not blurred.
5. They must be **close together** (d small) and the screen far enough away, so the fringes are wide enough to see.
6. The waves should have the **same state of polarisation**.
7. The path difference must be smaller than the **coherence length**.

CHECK YOURSELF

Two monochromatic waves, each of intensity I, completely incoherent. Resultant intensity?

$2I$ everywhere (no fringes), because $\langle \cos\phi \rangle = 0$.

Wave trains last 10^{-10} s. Coherence length?

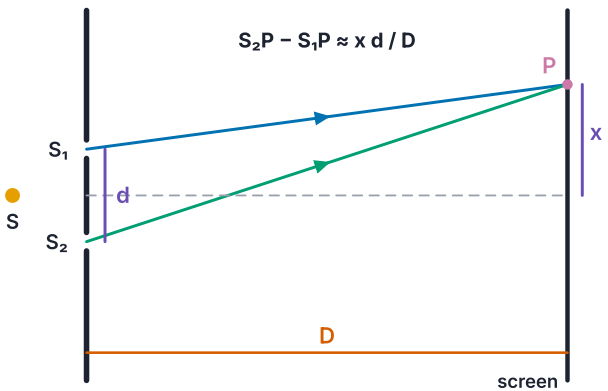
$3 \times 10^{8} \times 10^{-10} = 0.03$ m $= 3$ cm.

In the model paper: **Q3** **Q4** **Q5** **Q6** **Q7** **Q18**.

Young's fringes, properly

T3 Path difference and fringe width

> Why the fringes are equally spaced (PYQ N25 Q14, N22 Q7).



Young's double-slit arrangement. Light from one slit S falls on two slits S_1, S_2 (distance d apart). They act as two coherent sources. On a screen at distance D, the point P at height x gets two waves with path difference $\Delta \approx xd/D$.

Slits S_1, S_2 are d apart; the screen is at distance D. P is a point on the screen at height x above the centre O.

1. $(S_2P)^2 = D^2 + \left(x + \frac{d}{2}\right)^2$, $(S_1P)^2 = D^2 + \left(x - \frac{d}{2}\right)^2$
   Pythagoras: horizontal side D, vertical side the height difference.

2. Subtract: $(S_2P)^2 - (S_1P)^2 = 2xd$
   D^2 cancels; $(x + \frac{d}{2})^2 - (x - \frac{d}{2})^2 = 2xd$.

3. $S_2P - S_1P = \dfrac{2xd}{S_2P + S_1P}$
   $p^2 - q^2 = (p - q)(p + q)$.

4. $D \gg d, x$, so $S_2P + S_1P \approx 2D$: $\Delta = S_2P - S_1P \approx \dfrac{xd}{D}$
   M3. For d = 0.02 cm, D = 50 cm, x = 0.5 cm the error is 0.005%.

5. Bright: $\dfrac{x_n d}{D} = n\lambda \Rightarrow x_n = \dfrac{n\lambda D}{d}$, $n = 0, 1, 2, \ldots$

6. Fringe width: $\beta = x_{n+1} - x_n = \dfrac{\lambda D}{d}$. Dark fringes at $x = (n + \frac{1}{2})\dfrac{\lambda D}{d}$, exactly halfway
   β does not depend on n: the fringes are equally spaced.

RESULT

$$\beta = \frac{\lambda D}{d}$$

Bigger λ or bigger D ⟹ wider fringes. Bigger d ⟹ narrower fringes.

EXAM TRAP · "THE SCREEN MUST BE CLOSE TO THE SOURCES" (T/F)

False. $\beta \propto D$. Bring the screen closer and the fringes crowd together until they cannot be seen. The formula also needs $D \gg d$.

EXAM TRAP

β is the distance from one **bright** fringe to the next bright one (or dark to dark), not bright to dark.

The true shape of the fringes

THE IDEA

A fringe is a set of points with the same $S_2P - S_1P$. In a plane, the points whose distances from two fixed points differ by a constant form a **hyperbola** (Plus One conic sections). In space they are hyperboloids. Near the centre of a distant screen ($y \ll D$) these hyperbolas are almost straight lines parallel to the slits: that is what we see.

RESULT · GHATAK 14.22

$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ with $b^2 = D^2 + \frac{1}{4}(d^2 - \Delta^2)$. The central fringe (Δ = 0) is the straight line $x = 0$.

CHECK YOURSELF

λ = 500 nm, d = 0.5 cm, D = 1 m. Fringe width?

$$\beta = \frac{5 \times 10^{-7} \times 1}{5 \times 10^{-3}} = 10^{-4}\ \text{m} = 0.1\ \text{mm}.$$

In the model paper: **Q8** **Q10** **Q39** .

T4 Intensity distribution

> The 10-mark derivation (PYQ N19 Q32, N24 Q20).

Two coherent waves reach P from S_1 and S_2. Their electric fields (nearly parallel, since $D \gg d$):

$$E_1 = E_{01}\cos\left(\tfrac{2\pi}{\lambda}S_1P - \omega t\right), \quad E_2 = E_{02}\cos\left(\tfrac{2\pi}{\lambda}S_2P - \omega t\right)$$

1. Total field $E = E_1 + E_2$; intensity $I = K\langle E^2\rangle$
   Superposition (T1); intensity ∝ field² averaged (P2). K is a constant.

2. $E^2 = E_1^2 + E_2^2 + 2E_1E_2$

3. $2E_1E_2 =$
   $E_{01}E_{02}\left\{\cos\left[\tfrac{2\pi}{\lambda}(S_2P - S_1P)\right] + \cos\left[2\omega t - \tfrac{2\pi}{\lambda}(S_1P + S_2P)\right]\right\}$
   Identity (c) of M1: product → sum.

1. Average over time: $\langle E_1^2 \rangle = \frac{1}{2}E_{01}^2$, $\langle E_2^2 \rangle = \frac{1}{2}E_{02}^2$; the $2\omega t$ term averages to 0

   P2: $\langle \cos^2 \rangle = \frac{1}{2}$, $\langle \cos \rangle = 0$. The first cos does not depend on t, so it stays.

2. Write $I_1 = \frac{1}{2}KE_{01}^2$, $I_2 = \frac{1}{2}KE_{02}^2$, $\delta = \frac{2\pi}{\lambda}(S_2P - S_1P)$:

RESULT · TWO-BEAM INTERFERENCE

$$I = I_1 + I_2 + 2\sqrt{I_1 I_2}\cos\delta$$

Maximum $\delta = 2m\pi$: $I_{\max} = (\sqrt{I_1} + \sqrt{I_2})^2$

Minimum $\delta = (2m+1)\pi$: $I_{\min} = (\sqrt{I_1} - \sqrt{I_2})^2$

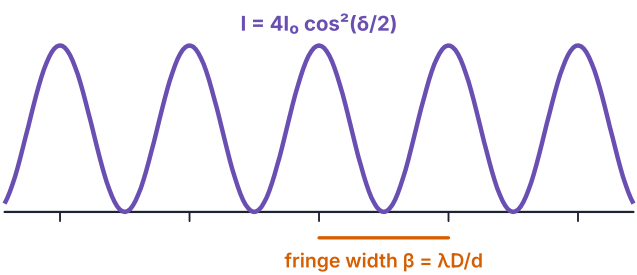
Equal beams

1. $I_1 = I_2 = I_0$: $I = 2I_0(1 + \cos\delta) = 4I_0\cos^2\frac{\delta}{2}$

   Identity (d) of M1.

2. Maximum $4I_0$, minimum 0, average $2I_0$

   Average of $\cos^2$ is ½. The average equals $I_1 + I_2$: energy is only moved around, not created.



Intensity on the screen. Bright fringes ($4I_0$) where $\delta = 0, 2\pi, \ldots$; dark (zero) where $\delta = \pi, 3\pi, \ldots$. Neighbouring bright fringes are β apart, all equally spaced.

Incoherent sources

If the phase difference δ changes at random, $\langle \cos\delta \rangle = 0$ and $I = I_1 + I_2$ (equal sources: $2I_0$) everywhere: uniform light, no fringes (T2).

Visibility (contrast) of fringes

RESULT

$$V = \frac{I_{\max} - I_{\min}}{I_{\max} + I_{\min}} = \frac{2\sqrt{I_1 I_2}}{I_1 + I_2}$$

$V = 1$ for equal beams (perfectly dark minima); $V \to 0$ when one beam is much weaker. This is why condition 3 of sustained interference asks for equal amplitudes.

SHORT WAY (EQUAL AMPLITUDES)

$a\cos\omega t + a\cos(\omega t + \delta) = 2a\cos\dfrac{\delta}{2}\cos(\omega t + \dfrac{\delta}{2})$ by identity (b). Amplitude $2a\cos\dfrac{\delta}{2}$; intensity $\propto 4a^2\cos^2\dfrac{\delta}{2}$.

CHECK YOURSELF

$I_1 = 4I_2$. Find $I_{\max}/I_{\min}$ and V.

$\sqrt{I_1} = 2\sqrt{I_2}$: $I_{\max} = 9I_2$, $I_{\min} = I_2$, ratio 9. $V = 8/10 = 0.8$.

In the model paper: **Q2** **Q3** **Q9** **Q39** .

T5 Other fringe shapes; wide sources

> Ghatak §14.6 examples. Not yet asked, but in the syllabus section.

Two plane waves crossing

Two plane waves of equal amplitude travel at angles +θ and −θ to the z-axis.

1. $E = E_0 \cos(kz\cos\theta + kx\sin\theta - \omega t) + E_0 \cos(kz\cos\theta - kx\sin\theta - \omega t)$
   A wave going at angle θ has phase $k(z\cos\theta \pm x\sin\theta) - \omega t$.

2. $E = 2E_0 \cos(kx\sin\theta)\cos(kz\cos\theta - \omega t)$
   Identity (b) of M1.

3. $I = 4I_0\cos^2(kx\sin\theta)$: straight fringes along z, spacing $\beta = \dfrac{\lambda}{2\sin\theta}$
   $\cos^2$ repeats when $kx\sin\theta$ grows by π, i.e. x grows by $\pi/(k\sin\theta) = \lambda/(2\sin\theta)$.

CHECK YOURSELF

θ = 30°, λ = 5000 Å. β?

$\beta = \lambda/(2\times 0.5) = \lambda = 5000$ Å $= 0.5$ μm.

Screen perpendicular to the line S_1S_2

If the screen faces along the line joining the two sources, Δ is the same on circles round the axis: the fringes are **circles** (this is the shape seen in Michelson, T20).

A wide source

THE IDEA

Each point of a wide source S makes its own fringe pattern, slightly shifted. The patterns of different points are incoherent, so their **intensities** add. If the source is too wide, bright fringes of one point fall on dark fringes of another, and the contrast is lost. This is why the first slit must be **narrow** (spatial coherence).

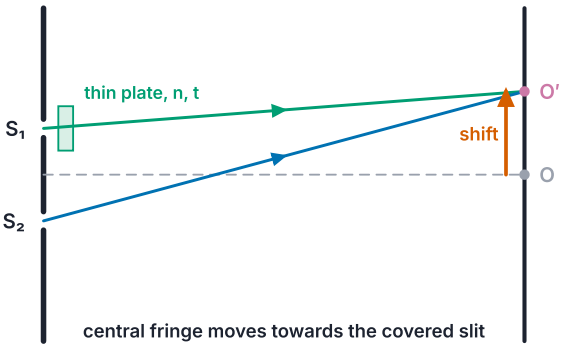
RESULT

Fringes stay visible only if the source width w satisfies roughly $w \lesssim \dfrac{\lambda D_s}{2d}$ (D_s = source-to-slit distance).

In the model paper: **Q32** ; also used in **Q4** .

T6 Shift of fringes by a thin plate

Ghatak §14.10. A favourite numerical.



A thin transparent plate over one slit adds an extra optical path $(n-1)t$ to that beam. The point of zero path difference (the central fringe) moves from O to O′, towards the covered slit, by $\Delta = D(n-1)t/d$.

1. Time from S_1 to P through a plate of thickness t:
$$\frac{S_1P - t}{c} + \frac{t}{v} = \frac{1}{c}[S_1P + (n-1)t]$$
$v = c/n$, so $t/v = nt/c$.

2. So the plate adds the optical path $(n-1)t$ to beam 1 (P4)

3. The new central fringe O′ is where both total paths are equal: $S_1O' + (n-1)t = S_2O'$

4. From T3: $S_2O' - S_1O' = \dfrac{d}{D}\,OO'$

5. So $(n-1)t = \dfrac{d}{D}\,OO'$, and the shift is $\Delta x = \dfrac{D(n-1)t}{d}$
Ghatak 14.37.

RESULT

$$\Delta x = \frac{D(n-1)t}{d} = \frac{(n-1)t}{\lambda}\,\beta$$

The whole pattern shifts towards the covered slit by $(n-1)t/\lambda$ fringes. The fringe width does not change.

EXAM TRAP

Use $(n-1)t$, not nt: the plate replaces a thickness t of air. The shift is **towards** the slit with the plate (that beam is "slowed", so it needs a shorter geometric path).

1. Example (Ghatak 14.10): mica $n = 1.58$, $d = 0.1$ cm, $D = 50$ cm, shift 0.2 cm. $t = \dfrac{d\,\Delta x}{D(n-1)} = \dfrac{0.1 \times 0.2}{50 \times 0.58}$

2. $t = 6.9 \times 10^{-4}$ cm
   The book prints 6.7; the arithmetic gives 6.9.

CHECK YOURSELF

After inserting the plate, the centre O is occupied by the 11th bright fringe (λ = 5893 Å, n = 1.58). t?

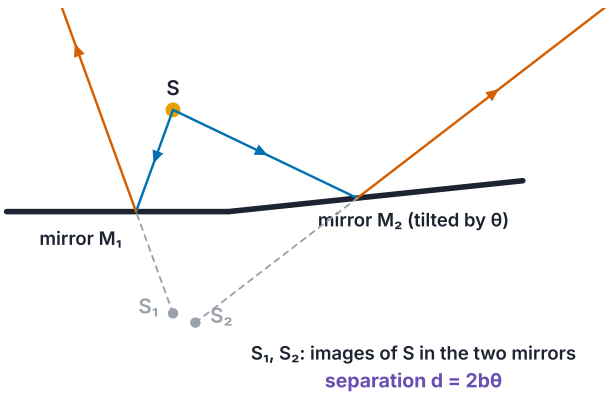
$(n-1)t = 11\lambda$: $t = 11 \times 5893 \times 10^{-8}/0.58 = 1.12 \times 10^{-3}$ cm.

In the model paper: **Q26** .

Two sources from one: mirrors and prisms

T7 Fresnel's two mirrors

> Division of wavefront with two mirrors (in the syllabus; not yet asked).



Fresnel's two-mirror arrangement. Two mirrors meet at a very small angle θ. The images S_1, S_2 of the source in the two mirrors are very close and coherent, like the two slits of Young.

THE IDEA

Two plane mirrors M_1, M_2 meet at a line M, tilted by a very small angle θ. A slit S sends light onto both. Each mirror forms an image of S: S_1 and S_2. The two reflected beams look as if they come from S_1 and S_2, two copies of one source: **coherent**. They overlap and give Young-type fringes.

1. $MS_1 = MS = MS_2 = b$
   A plane-mirror image is as far behind the mirror as the object is in front; the meeting line M lies on both mirrors, so M is the same distance from S and from each image.

2. So S, S_1 and S_2 lie on a circle with centre M and radius b

3. $\angle S_1 M S_2 = 2\theta$
   Turning the mirror by θ turns the image about M by 2θ (P5).

4. $d = S_1S_2 = b \cdot 2\theta = 2b\theta$
   Arc ≈ chord for a small angle: radius × angle (M3).

5. With the screen at distance c from M, $D = b + c$ and $\beta = \dfrac{\lambda D}{d} = \dfrac{\lambda(b+c)}{2b\theta}$

RESULT

$$d = 2b\theta, \qquad \beta = \frac{\lambda(b+c)}{2b\theta}$$

EXAM TRAP

$S_1S_2 = 2b\theta$, not $b\theta$; and $D = b + c$, not just c.

CHECK YOURSELF

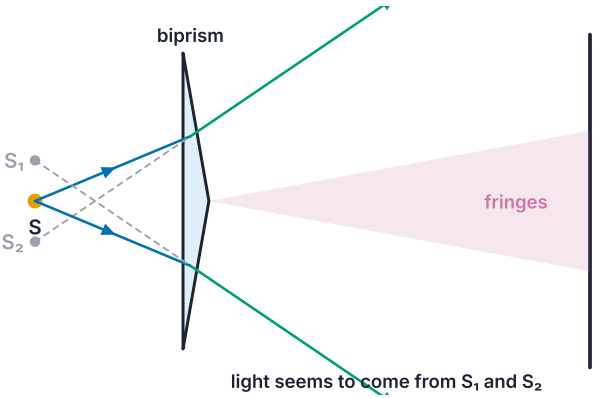
Why are S_1 and S_2 coherent?

They are images of the same slit S, so every phase jump of S happens in both at once.

In the model paper: **Q12** .

T8 Fresnel's biprism

How it works and how it measures λ (PYQ N20 Q19, N19 Q26).



Fresnel's biprism. Each half of the thin prism bends the light slightly. Behind it the light seems to come from two virtual sources S_1 and S_2 (grey dashed). They are copies of one source, so they are coherent. Fringes form where the two beams overlap (pink).

THE IDEA

A biprism is two thin prisms joined base to base (obtuse angle nearly 180°; each refracting angle α is tiny, about 30′). Light from a slit S falling on the upper half is bent down; on the lower half, bent up. Seen from the right, the light seems to come from two virtual sources S_1 and S_2, copies of S: **coherent**.

1. Each half bends rays by $\delta = (n-1)\alpha$
   Thin prism (P5).

2. With the slit at distance a from the biprism, each virtual source is shifted sideways by $a\delta = a(n-1)\alpha$
   Small angle: sideways shift = distance × angle (M3).

3. So $d = S_1S_2 = 2a(n-1)\alpha$

4. Screen (eyepiece) at distance b beyond the biprism: $D = a + b$, so $\beta = \dfrac{\lambda(a+b)}{2a(n-1)\alpha}$

RESULT

$$d = 2a(n-1)\alpha, \quad \beta = \frac{\lambda D}{d}, \quad \lambda = \frac{\beta\, d}{D}$$

1. Example (PYQ N19 Q26): $n = 1.5$, α = 1° $= 0.01745$ rad, $a = 10$ cm, $b = 100$ cm, λ = 5000 Å

2. $d = 2 \times 10 \times 0.5 \times 0.01745 = 0.1745$ cm; $D = 110$ cm

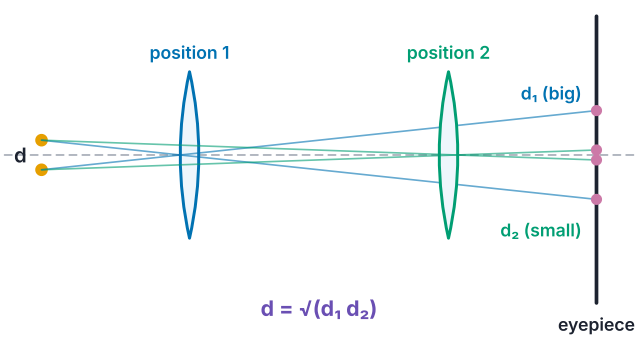
3. $\beta = \dfrac{5 \times 10^{-5} \times 110}{0.1745} = 0.0315$ cm ≈ 0.32 mm

Measuring λ with a biprism

THE SET-UP

Slit, biprism and micrometer eyepiece stand on an optical bench. The slit is lit by sodium light and made parallel to the edge of the biprism, so fringes appear in the eyepiece.

1. **β**: move the eyepiece cross-wire across, say, 20 fringes and read the micrometer; β = distance ÷ 20
2. **D**: read from the bench scale (slit to eyepiece)
3. **d**: lens displacement method (below)
4. $\lambda = \beta d / D$



Lens displacement method for the source separation d. A convex lens between the biprism and the eyepiece gives sharp images of S_1, S_2 at two positions. Magnifications are v/u and u/v, so $d_1 = d \cdot v/u$, $d_2 = d \cdot u/v$ and $d_1 d_2 = d^2$. (Needs D > 4f.)

Why d = √(d₁d₂)

1. Keep slit and eyepiece fixed, distance D apart. A convex lens of focal length f placed between them forms a sharp image when $u + v = D$ and the lens formula holds
   Lens formula (P5), with distances as magnitudes.
2. This gives a quadratic, $u^2 - Du + Df = 0$, which has two real roots only if $D^2 \geq 4Df$, i.e. $D \geq 4f$
   So choose a lens with $f < D/4$.
3. The two positions swap u and v. Image separations: $d_1 = d\dfrac{v}{u}$, $d_2 = d\dfrac{u}{v}$
   Magnification = v/u.
4. Multiply: $d_1 d_2 = d^2 \Rightarrow d = \sqrt{d_1 d_2}$

EXAM TRAP

$d = \sqrt{d_1 d_2}$ (geometric mean), not $(d_1 + d_2)/2$. In $d = 2a(n-1)\alpha$, a is the slit-to-biprism distance, and α must be in radians.

CHECK YOURSELF

d_1 = 0.6 cm, d_2 = 0.15 cm. d?

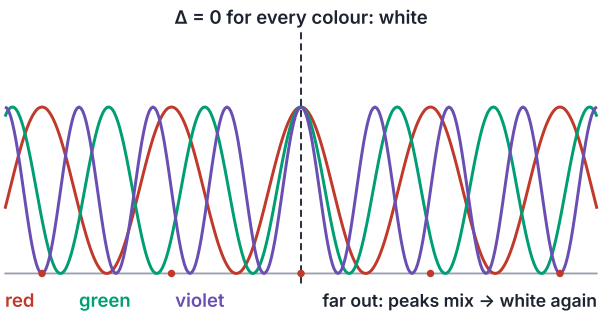
$\sqrt{0.09} = 0.3$ cm.

In the model paper: **Q27** **Q28**.

White light, Lloyd, and a half-wave jump

T9 Interference with white light

> Why you see only a few coloured fringes.



White-light fringes. Each colour makes its own fringes with its own width β = λD/d (red widest). All are bright at the centre, so the central fringe is white. A few coloured fringes follow; further out the colours overlap and the screen looks uniformly white.

1. At the centre O, $\Delta = 0$ for every wavelength: all colours are bright there $\Longrightarrow$ the central fringe is **white**
2. Moving out, Δ grows. At $\Delta \approx 2 \times 10^{-5}$ cm (half the violet wavelength) violet is cancelled: the edge looks reddish
3. At $\Delta \approx 3.5 \times 10^{-5}$ cm (half the red wavelength) red is cancelled: the next region looks bluish. So the first few fringes are **coloured**, violet on the inside, red on the outside
4. Far out (say $\Delta = 30 \times 10^{-5}$ cm) many different wavelengths are bright ($\Delta = m\lambda$) and many dark at the same point. The mixture looks white again: uniform illumination

   The fringes of different colours overlap and wash out.

RESULT

White light gives a **white central fringe**, then a few coloured fringes, then uniform white. Use: the white central fringe marks exactly where $\Delta = 0$ (to locate the zero-order fringe, for example when measuring a fringe shift, T6).

CHECK YOURSELF

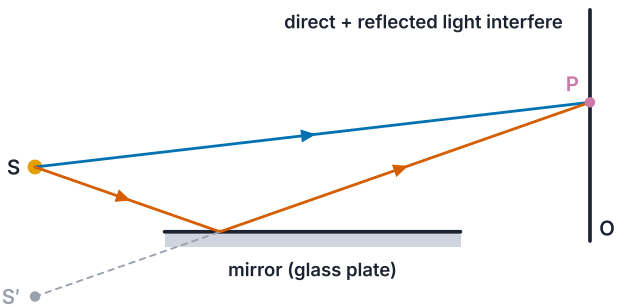
At a point where $\Delta = 5000$ Å, which visible colour is brightest?

Only $\lambda = 5000$ Å (green) has $\Delta = m\lambda$ in the visible; the point looks greenish.

In the model paper: **Q11** .

T10 Lloyd's mirror

> One real source, its mirror image as the second (PYQ N22 Q26).



Lloyd's mirror. Light reaches P directly and after reflection at grazing incidence, as if from the virtual source S′. At O, where the screen touches the mirror's plane, the paths are equal, yet the fringe is **dark**: reflection added a phase change of π.

THE IDEA

A slit S is placed a small height h above a long flat glass plate (the mirror). Light grazing the mirror reflects and seems to come from the image S′, a distance h below the mirror. S and S′ are coherent, $d = 2h$ apart. Direct and reflected light overlap near the mirror and give fringes.

1. Fringe width as in Young: $\beta = \dfrac{\lambda D}{2h}$
   d = 2h; D = source-to-screen distance.
2. Expected: where the screen touches the mirror (O), the paths are equal, so a bright fringe
3. Observed: the fringe at O is **dark**. Every bright and dark fringe is swapped
   Seen by moving the screen up to the end of the mirror.
4. Conclusion: reflection at the glass (denser) surface adds a phase change of π (half a wave) to the reflected wave

RESULT · LLOYD'S MIRROR

Dark	$S'P - SP = m\lambda$
Bright	$S'P - SP = (m + \frac{1}{2})\lambda$
Width	$\beta = \lambda D/2h$

Central fringe dark. The fringes appear only on one side of the mirror plane.

1. Example (PYQ N22 Q26): h = 2 mm, so d = 4 mm; λ = 589 nm; D = 1.5 m
2. $\beta = \dfrac{589 \times 10^{-9} \times 1.5}{4 \times 10^{-3}} = 2.21 \times 10^{-4}$ m $= 0.221$ mm

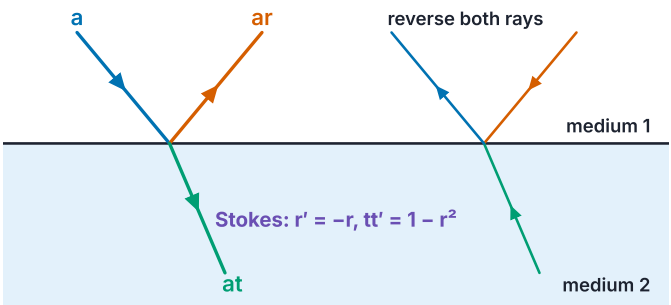
EXAM TRAP

d = 2h, not h. Lloyd's centre is **dark**, opposite to Young.

In the model paper: **Q13** **Q29** .

T11 Phase change on reflection: Stokes relations

> Why only one of the two reflections gets the half-wave jump.



Stokes relations. Reverse the reflected and transmitted rays: they must rebuild the original ray only. This forces $r' = -r$: reflection from one side differs by a phase of π from reflection from the other side.

THE IDEA · REVERSIBILITY

If you reverse a ray (send it back along its path) it retraces its path exactly. Stokes applied this to a ray that splits into a reflected and a transmitted part.

Let r, t be the amplitude ratios for reflection and transmission from medium 1, and r', t' from medium 2.

1. A wave of amplitude a in medium 1 gives a reflected wave ar and a transmitted wave at
2. Reverse both. The reversed ar splits into $ar \cdot r$ (back along the original incident line) and $ar \cdot t$ (into medium 2). The reversed at splits into $at \cdot t'$ (into medium 1) and $at \cdot r'$ (into medium 2)
   Each reversed wave hits the surface and divides again.
3. Reversibility says the result must be just the original wave a, travelling backwards in medium 1
4. Medium 1: $ar^2 + att' = a \implies tt' = 1 - r^2$
5. Medium 2 (nothing should emerge): $art + atr' = 0 \implies r' = -r$

RESULT · STOKES RELATIONS

$$r' = -r, \qquad tt' = 1 - r^2$$

The minus sign = a phase change of π. So the reflections from the two sides of a surface differ by π. Lloyd's mirror shows it is the reflection from the **denser** medium that gets the π; the reflection from the rarer side gets none.

SAME RESULT FROM FRESNEL'S FORMULAS (NORMAL INCIDENCE)

$r = \dfrac{n_1 - n_2}{n_1 + n_2}$, $t = \dfrac{2n_1}{n_1 + n_2}$. Swapping n_1, n_2: $r' = \dfrac{n_2 - n_1}{n_1 + n_2} = -r$, $t' = \dfrac{2n_2}{n_1 + n_2}$.

Check: $1 - tt' = 1 - \dfrac{4n_1 n_2}{(n_1 + n_2)^2} = \dfrac{(n_1 - n_2)^2}{(n_1 + n_2)^2} = r^2$ ✓. And $r < 0$ when $n_2 > n_1$: the π change on reflection from a denser medium.

RESULT · THE RULE USED IN ALL THIN FILMS

Rare → dense	reflected wave gets π (adds λ/2)
Dense → rare	no phase change
Transmitted	never a phase change

CHECK YOURSELF

Light in air reflects from glass (n = 1.5) at normal incidence. Fraction of intensity reflected?

$r = -0.5/2.5 = -0.2$; $r^2 = 0.04$: 4%, with a phase change of π.

In the model paper: **Q13** **Q30**.

Splitting the amplitude: thin films

T12 A thin film at normal incidence

> The simplest film, and the half-wave rule (PYQ N19 Q13, N19 Q5, N22 Q27).

THE IDEA

Light falls straight down on a thin transparent film (a soap film, an oil layer) of index n and thickness t, with air on both sides. Part reflects at the top surface (ray 1). Part enters, reflects at the bottom, and comes back out (ray 2). These two are copies of the same wave: **division of amplitude**.

1. Ray 2 goes down and up through the film: extra optical path $2nt$
   Optical path = n × length (P4).
2. Ray 1 reflects air → film (rare → dense): phase change π. Ray 2 reflects film → air (dense → rare): none
   T11.
3. So the two rays differ by $2nt$ plus an extra half wave

RESULT · FILM IN AIR, REFLECTED LIGHT, NORMAL INCIDENCE

Dark $2nt = m\lambda$

Bright $2nt = (m + \frac{1}{2})\lambda$

$m = 0, 1, 2, \ldots$ In **transmitted** light the conditions are swapped (no reflection π difference), so a film that looks bright in reflection looks dark in transmission, and vice versa.

When the half-wave rule changes

COUNT THE Π'S

What matters is whether the two rays get a **different** number of π changes.

- Film in air, or air film between glass plates: one π (one reflection rare → dense) ⟹ the conditions above.
- $n_1 < n_2 < n_3$ (for example oil 1.60 between crown glass 1.52 above and flint 1.66 below, or a coating on glass): **both** reflections are rare → dense, both get π, which cancel ⟹ conditions swap: bright $2nt = m\lambda$, dark $2nt = (m + \frac{1}{2})\lambda$.

EXAM TRAP · FILL IN THE BLANK

"The optical path of light reflected from the lower surface of a thin film **increases** with increase of refractive index of the film." Because the extra path is $2nt$.

1. Example (PYQ N22 Q27): soap bubble n = 1.42, λ = 5839 Å. Least thickness that looks bright in reflection?
2. Bright: $2nt = (m + \frac{1}{2})\lambda$; least thickness for $m = 0$: $t = \frac{\lambda}{4n}$
3. $t = \frac{5839}{4 \times 1.42} = 1028$ Å $\approx 1.03 \times 10^{-7}$ m

REMIND ME

Inside the film the wavelength is λ/n. So $2nt = m\lambda$ just says "the round trip $2t$ holds m wavelengths of the film".

CHECK YOURSELF

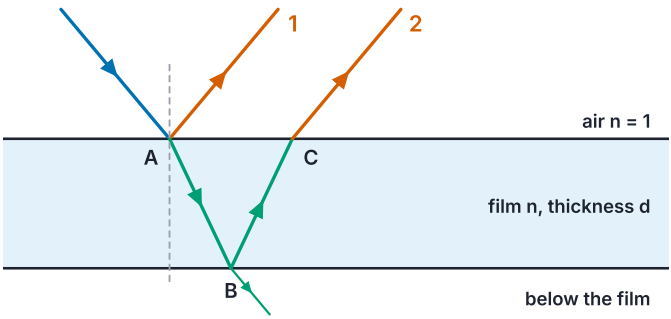
A very thin soap film (t → 0) in reflected light looks...?

Dark: $2nt = 0 = m\lambda$ with m = 0. Only the π difference is left, and the rays cancel. (Soap films look black just before they burst.)

In the model paper: **Q14** **Q15** **Q16** .

T13 Oblique incidence: the cosine law

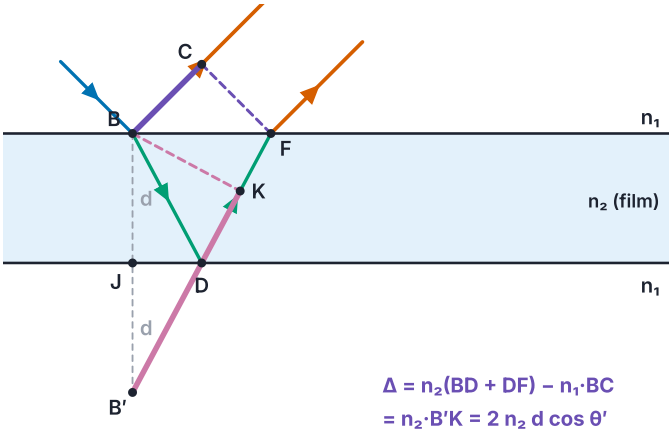
> The 5- and 10-mark derivation (PYQ N24 Q14, N25 Q20, N24 Q13).



path difference of rays 1 and 2: Δ = 2 n d cos r

A thin film. Ray 1 reflects at the top surface; ray 2 goes into the film, reflects at the bottom and comes out. Ray 2 travels the extra optical path $2nd\cos r$ (cosine law), where r is the angle inside the film.

A plane wave falls at angle θ on a film of index n_2 and thickness d, in a medium of index n_1. Inside, the angle is θ′ with $n_1 \sin\theta = n_2 \sin\theta'$.



Cosine-law construction. One ray splits at B: reflected along BC (orange), refracted along B→D→F. FC ⊥ the reflected rays (outgoing wavefront). B′ is the mirror image of B in the lower surface (BJ = JB′ = d), so BD = B′D and B′, D, F lie on one line. BK ⊥ B′F. Snell's law makes $n_1 \cdot BC$ (purple, thick) = $n_2 \cdot KF$, which leaves the extra path $n_2 \cdot B'K$ (pink, thick).

1. One incident ray splits at B: part reflects along BC, part enters, reflects at D and leaves at F, parallel to the first
   Division of amplitude: both come from the same ray.

2. Draw FC ⊥ the two outgoing rays (an outgoing wavefront). Beyond C and F the two rays travel equal paths, so $\Delta = n_2(BD + DF) - n_1 BC$
   Both start together at B; one goes B → C outside, the other B → D → F inside.

1. Let B′ be the mirror image of B in the lower surface (BJ = JB′ = d). Then $BD = B'D$, and B′, D, F are on one straight line

   Reflection at D is like coming straight from the image point (P5). So $BD + DF = B'F$.

2. $\angle CFB = \theta$, so $BC = BF \sin\theta$

   FC ⊥ the reflected ray, which makes angle θ with the normal; so FC makes angle θ with the surface. In the right triangle BCF, BC is opposite that angle.

3. Drop BK ⊥ B′F. B′F makes angle θ′ with the normal, so $90^\circ - \theta'$ with the surface; in the right triangle BKF, $KF = BF\cos(90^\circ - \theta') = BF\sin\theta'$

4. Snell: $n_1 \sin\theta = n_2 \sin\theta' \Rightarrow n_1\, BC = n_1 BF \sin\theta = n_2 BF \sin\theta' = n_2\, KF$

5. $\Delta = n_2\, B'F - n_2\, KF = n_2\, B'K$

6. In triangle BKB′: $BB' = 2d$ and $\angle KB'B = \theta'$, so $B'K = 2d\cos\theta'$

   B′F makes angle θ′ with the normal BB′.

RESULT · COSINE LAW

$$\Delta = 2n_2\, d\cos\theta'$$

θ′ is the angle of **refraction** (inside the film). Normal incidence: θ′ = 0, Δ = 2nd.

RESULT · MAXIMA AND MINIMA (FILM IN AIR, $n_2 = n$)

Reflected, bright	$2nd\cos\theta' = (m + \frac{1}{2})\lambda$
Reflected, dark	$2nd\cos\theta' = m\lambda$
Transmitted, bright	$2nd\cos\theta' = m\lambda$
Transmitted, dark	$2nd\cos\theta' = (m + \frac{1}{2})\lambda$

Reflected and transmitted patterns are **complementary** (energy is conserved).

EXAM TRAP

Use cos θ′ (inside angle), not cos θ. If asked for i: $\cos\theta' = \sqrt{1 - \sin^2 i/n^2}$, so $\Delta = 2d\sqrt{n^2 - \sin^2 i}$.

Why the reflected fringes are sharp, the transmitted faint

Ghatak's example (n = 1.36): the two main reflected rays have amplitudes 0.153a and 0.149a (almost equal ⟹ near-perfect dark fringes). The two main transmitted rays have 0.977a and 0.023a (very unequal ⟹ poor contrast; see visibility in T4).

A film lit by a plane wave

THE IDEA · PYQ N24 Q13

In a plane wave every ray has the same θ, so θ′ is the same everywhere. If the film has the same thickness everywhere, Δ is the same at every point: the whole film looks **uniformly** bright, or uniformly dark, or in between. There are no fringes.

If the thickness varies a little, bright and dark bands follow lines of equal thickness (wedge, T17; Newton's rings, T18).

Looking at a large film, the eye's pupil collects reflected light only from a small region, so it sees only a small part at a time.

If the film is too thick, Δ exceeds the coherence length (T2): no fringes.

CHECK YOURSELF

n = 1.5, d = 2000 Å, θ′ = 60°. Δ?

$2 \times 1.5 \times 2000 \times 0.5 = 3000$ Å.

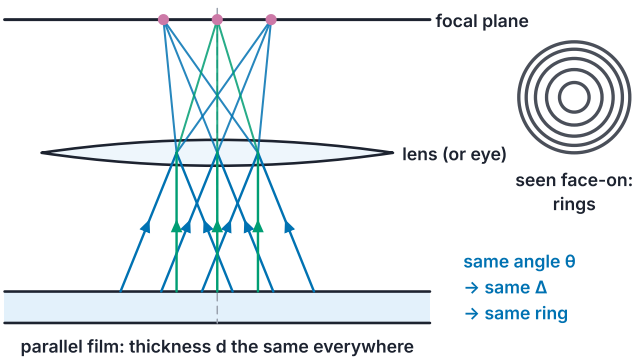
In the model paper: **Q31** **Q40**.

T14 Point or extended source: Haidinger's fringes

Fringes of equal inclination (PYQ N24 Q16, N20 Q20).

THE IDEA

From a point source, rays reach a parallel film at **different angles**. Δ = 2nd cos θ′ now changes with the angle. All rays with the same angle lie on a cone around the normal from the source, so the fringes are **circles**.



Haidinger's fringes (fringes of equal inclination). The film thickness is fixed; only the angle changes. All rays leaving the film at the same angle θ (blue, on either side) have the same path difference 2nd cos θ′ and are focused on one circle in the focal plane. So each fringe is a ring, seen focused at infinity.

RESULT · HAIDINGER'S FRINGES

- **Conditions**: a film (plate) of **constant** thickness; a broad (extended) monochromatic source; viewing with the eye or a telescope **focused at infinity**, near normal incidence.
- Each fringe = one value of θ′: **fringes of equal inclination**, concentric rings, localised at infinity.
- At the centre θ′ = 0, cos θ′ largest, so the **order is highest at the centre** and decreases outward.

	HAIDINGER	NEWTON'S RINGS
Film	constant thickness	thickness varies (air gap under a lens)
Fringe =	same angle	same thickness
Seen at	infinity (focus far)	the film itself
Order	highest at centre	lowest (0) at centre

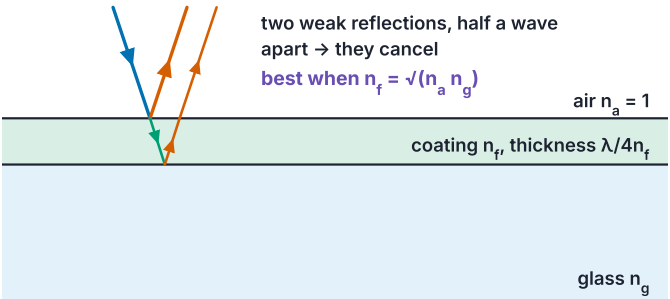
LOCALISATION

With an extended source, fringes are clear only in one place. For a parallel film: at infinity. For a wedge (T17) or Newton's rings: at (very near) the film, so you must focus on the film. With a point source the fringes are seen everywhere (not localised).

In the model paper: **Q33**.

T15 Non-reflecting (anti-reflection) films

Why camera lenses look purple (PYQ N19 Q22, N24 Q15, N22 Q4).



A non-reflecting (anti-reflection) film. Both reflections happen at a "rarer → denser" boundary, so both get the same phase change, which cancels out. The quarter-wave thickness makes them half a wave apart, so they cancel.

THE PROBLEM

Each air–glass surface reflects about 4% of the light (T11 check). A camera with many lens surfaces loses a lot of light and gets stray reflections (flare). Fix: coat the lens with a thin transparent film that makes the two reflections cancel.

Air n_a, film n_f, glass n_g, with $n_a < n_f < n_g$ (for example MgF_2, $n_f = 1.38$, on glass 1.5).

1. Both reflections are rare → dense: both get π, which cancel
   T12, "count the π's".

2. For cancellation, the path difference must be half a wave: $2n_f d = \frac{\lambda}{2} \Longrightarrow d = \frac{\lambda}{4n_f}$
   A "quarter-wave" film: a quarter of the wavelength inside the film.

3. For **complete** cancellation the two reflected amplitudes must be equal: $\frac{n_f - n_a}{n_f + n_a} = \frac{n_g - n_f}{n_g + n_f}$
   Fresnel's r (T11); the transmission factors through the top surface are close to 1 (about 0.97).

4. Cross-multiply and simplify: $n_f^2 = n_a n_g \Longrightarrow n_f = \sqrt{n_a n_g}$

RESULT · NON-REFLECTING FILM

$$d = \frac{\lambda}{4 n_f}, \qquad n_f = \sqrt{n_a\, n_g}$$

Ideal for glass 1.5	$n_f = 1.22$ (no hard material that low; MgF_2 1.38 is used)
MgF_2, λ = 5000 Å	$d = 5000/(4 \times 1.38) \approx 900$ Å
Reflection	4% uncoated → about 1.4% with MgF_2

WHERE DOES THE LIGHT GO?

Not destroyed: the energy not reflected goes into the **transmitted** beam. The lens passes more light.

WHY PURPLE

The film is a quarter-wave only for one wavelength, chosen in the middle of the spectrum (yellow-green, where the eye is most sensitive). Red and violet are not fully cancelled and are partly reflected: red + violet = purple tint.

EXACT RESULT (GHATAK 15.30–15.36)

$R = \dfrac{r_1^2 + r_2^2 + 2 r_1 r_2 \cos 2\delta}{1 + r_1^2 r_2^2 + 2 r_1 r_2 \cos 2\delta}$, $\delta = \dfrac{2\pi n_f d}{\lambda}$. Minimum at $d = \dfrac{\lambda}{4 n_f}, \dfrac{3\lambda}{4 n_f}, \ldots$ with $R_{\min} = \left(\dfrac{n_a n_g - n_f^2}{n_a n_g + n_f^2} \right)^2$. The thinnest (λ/4) is preferred: its dip is broadest over wavelengths.

In the model paper: **Q19** **Q34**.

Colours, wedges, rings and Michelson

T16 Colours of thin films

> Asked in four different years (PYQ N19 Q14, N22 Q14, N24 Q7, N22 Q8, N19 Q17).

THE IDEA

White light falls on a soap bubble or an oil film on water. For a given thickness and viewing angle, $2nd\cos\theta'$ is fixed. Some wavelengths satisfy the **bright** condition $2nd\cos\theta' = (m+\frac{1}{2})\lambda$ and are reflected strongly; others satisfy the **dark** condition and are missing. The film shows the mixture of the reinforced colours.

As the thickness changes from place to place (and with the viewing angle), the colour changes: coloured bands following lines of equal thickness.

EXAM TRAP · "COLOURS OF SOAP BUBBLES ARE DUE TO DISPERSION" (T/F)

False. They are due to **interference** between light reflected from the two surfaces of the film. (Dispersion = splitting by a prism, because n depends on λ.)

No colours in thick plates

RESULT · TWO REASONS

1. In a thick plate, $2nd$ is many wavelengths. At any point, very many wavelengths across the spectrum are bright and very many are dark; their mixture looks white (as far out in white-light fringes, T9).
2. The path difference is longer than the coherence length of white light (T2), so the two reflected beams are not coherent.

So colours appear only for films a few wavelengths thick.

CHECK YOURSELF

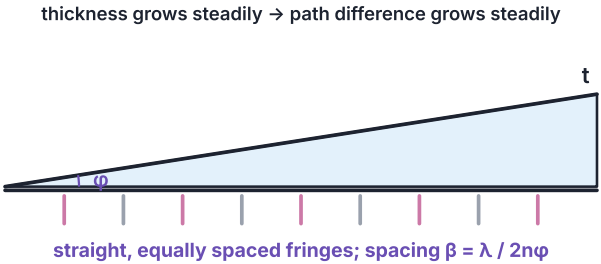
Why does a soap film's colour change as it drains?

Its thickness changes, so a different set of wavelengths satisfies the bright condition.

In the model paper: Q17 Q18 .

T17 The wedge-shaped film

> Straight, equally spaced fringes (PYQ N20 Q31, N20 Q10).



Wedge-shaped film (two plates touching at one edge, with a very small angle φ). Each fringe follows a line of equal thickness, so the fringes are straight lines parallel to the edge.

Two flat glass plates touch at one edge; a thin paper or wire at the other edge makes a thin wedge of air (or liquid of index n) with a very small angle φ. Light falls normally.

1. At distance x from the edge, the thickness is $t = x\tan\phi \approx x\phi$
   Small angle (M3).

2. Bright (reflected): $2nt = (m + \frac{1}{2})\lambda$, so $2n\phi\, x_m = (m + \frac{1}{2})\lambda$
   One π change, as in T12.

3. Next bright fringe: $2n\phi\, x_{m+1} = (m + \frac{3}{2})\lambda$

4. Subtract: $2n\phi(x_{m+1} - x_m) = \lambda \Rightarrow \beta = \frac{\lambda}{2n\phi}$

RESULT · WEDGE

$$\beta = \frac{\lambda}{2n\phi}$$

- Fringes are **straight** and **parallel to the edge** (each = a line of equal thickness), and **equally spaced** (β does not depend on m).
- From one fringe to the next, the thickness changes by $\lambda/2n$.
- At the edge (t = 0): **dark** in reflected light.

1. Example (PYQ N20 Q31): air wedge, λ = 6000 Å, 100 fringes per cm ⟹ β = 0.01 cm

2. $\phi = \frac{\lambda}{2\beta} = \frac{6 \times 10^{-5}}{2 \times 0.01} = 3 \times 10^{-3}$ rad $\approx 0.17^\circ$

EXAM TRAP · "FRINGES FROM TWO PERFECTLY PLANE SURFACES ARE NOT EQUAL" (T/F)

False. Between two optically flat plates the thickness grows evenly, so the fringes are equally spaced straight lines. Curved or uneven fringes show the surface is not flat (a test of flatness).

USES

Measuring the thickness of a thin wire, hair or foil (lab experiment 7): if it is at distance L from the edge, its thickness is $L\phi = L\lambda/2\beta$. Testing optical flatness.

CHECK YOURSELF

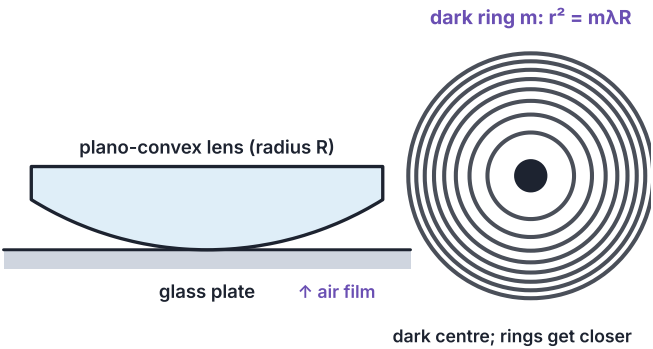
Air replaced by a liquid of n = 1.33. Fringe width?

Divided by 1.33: the fringes come closer.

In the model paper: Q20 Q35.

T18 Newton's rings

> The most-asked topic of the module (PYQ N22 Q32, N22 Q28, N19 Q27, N20 Q26, N19 Q3, N20 Q3).



Newton's rings. A curved lens sits on a flat plate, trapping an air film whose thickness grows with distance from the centre. Equal thickness lies on circles, so the fringes are rings. The centre is dark in reflected light.

THE SET-UP

A plano-convex lens of large radius R (curved side down) rests on a flat glass plate. Between them is a thin air film: zero thickness at the contact point, growing outward. Sodium light is turned down onto it by a glass plate at 45°, and the rings are viewed from above with a travelling microscope.

1. At distance r from the contact point the air film has thickness t, with $r^2 \approx 2Rt$
   Circle geometry (P6).

2. Light falls normally; air film (n = 1) between glass: one π change (glass → air at the lens: none; air → glass at the plate: π)
   T12, "count the π's".

3. Dark: $2t = m\lambda$. Put $2t = r^2/R$: $r_m^2 = m\lambda R$
   With a liquid of index n in the gap: $2nt = m\lambda$, so $r_m^2 = m\lambda R/n$.

4. Bright: $2t = (m + \frac{1}{2})\lambda \Longrightarrow r^2 = (m + \frac{1}{2})\lambda R$

5. Points of equal thickness lie on circles round the contact point ⟹ the fringes are **rings** (fringes of equal thickness)

RESULT · NEWTON'S RINGS (REFLECTED)

$$r_m^2 = m\lambda R \ (\text{dark}), \qquad D_m^2 = 4m\lambda R$$

- Centre (t = 0, m = 0): **dark** in reflected light; **bright** in transmitted light (complementary, but low contrast).
- $r \propto \sqrt{m}$: rings get **closer** further out (1, 1.41, 1.73, 2, ...).

Measuring λ

1. Measure the diameters of many dark rings with the travelling microscope (cross-wire on the left edge, then the right edge)
2. $D_{m+p}^2 = 4(m+p)\lambda R,\ \ D_m^2 = 4m\lambda R$
3. Subtract: $\lambda = \dfrac{D_{m+p}^2 - D_m^2}{4pR}$

   Using a difference cancels any error from dust or imperfect contact at the centre (which adds a constant to every D^2), and we need not know which ring is m = 1.
4. Find R with a spherometer. Or plot D^2 against m: a straight line of slope $4\lambda R$

RESULT

$$\lambda = \frac{D_{m+p}^2 - D_m^2}{4pR}, \qquad n_{\text{liquid}} = \frac{D_{\text{air}}^2}{D_{\text{liquid}}^2}$$

EXAM TRAP

The formula uses **diameters squared**. If given radii, $D^2 = 4r^2$. With a liquid the rings **shrink** by $\sqrt{n}$.

1. Example (PYQ N22 Q28): λ = 670 nm, $r_{20} = 1.1 \times 10^{-2}$ m. $R = \dfrac{r_{20}^2}{20\lambda} = \dfrac{1.21 \times 10^{-4}}{20 \times 6.7 \times 10^{-7}} = 9.03$ m

2. $r_{30} = r_{20}\sqrt{30/20} = 1.1 \times 10^{-2} \times 1.2247 = 1.35 \times 10^{-2}$ m

1. Example (PYQ N19 Q27): $D_{10} = 0.3$ cm, λ = 6×10^{-5} cm. $r_{10} = 0.15$ cm: $R = \dfrac{0.0225}{10 \times 6 \times 10^{-5}} = 37.5$ cm

2. Air film there: $t = \dfrac{m\lambda}{2} = \dfrac{10 \times 6 \times 10^{-5}}{2} = 3 \times 10^{-4}$ cm

EXAM TRAP · "9TH BRIGHT RING"

Bright rings: $r^2 = (m + \frac{1}{2})\lambda R$ with m = 0 for the **first** bright ring. So the 9th bright ring has m = 8. State the counting you use. (PYQ N20 Q26: R = 100 cm, λ = 6×10^{-5} cm gives 0.226 cm and 0.239 cm for the 9th and 10th.)

CHECK YOURSELF

Air replaced by water (n = 1.33). The 10th ring's diameter changes by what factor?

$1/\sqrt{1.33} = 0.867$: it shrinks by about 13%.

In the model paper: **Q21** **Q22** **Q36** **Q41**.

T19 Two close wavelengths: rings fade and return

Sodium light is really two lines, 5890 Å and 5896 Å (Ghatak §15.10).

THE IDEA

Each wavelength makes its own set of rings. Near the centre they coincide. Further out, the slightly shorter wavelength gets ahead; where its dark ring lands on a bright ring of the other, the rings blur out. Further still they come back into step.

1. Rings vanish where the orders differ by a half: $\frac{2t}{\lambda_2} - \frac{2t}{\lambda_1} = \frac{1}{2}$ ($\lambda_1 > \lambda_2$)
2. $2t\,\frac{\lambda_1 - \lambda_2}{\lambda_1 \lambda_2} = \frac{1}{2} \Rightarrow t_0 = \frac{\lambda_1 \lambda_2}{4(\lambda_1 - \lambda_2)}$
3. They reappear when the difference is 1: $t = 2t_0$. Then vanish again at $3t_0$, and so on

1. Sodium: $t_0 = \frac{5890 \times 5896 \times 10^{-16}}{4 \times 6 \times 10^{-8}}$ cm $= 1.45 \times 10^{-2}$ cm
2. That is near ring number $m \approx 2t_0/\lambda \approx 490$. With R = 100 cm, $r = \sqrt{2Rt_0} = 1.7$ cm

CHECK YOURSELF

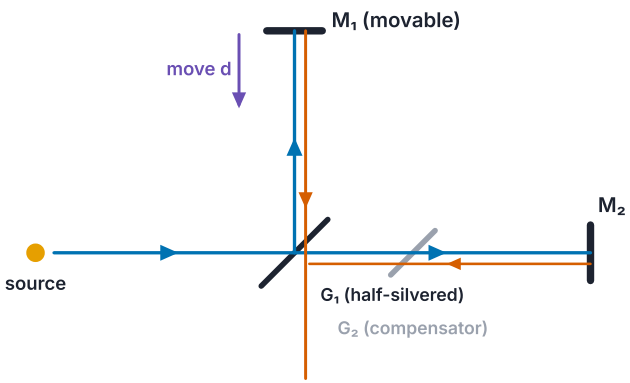
Why do Newton's rings in white light show only a few coloured rings?

Same idea with very many wavelengths: they fall out of step after a few rings, and the mixture looks white.

In the model paper: **Q37**.

T20 Michelson interferometer

> Construction, fringes, λ, the metre (PYQ N19 Q21, N20 Q32, N25 Q4).



Michelson interferometer. G_1 splits the beam into two (division of amplitude). After reflection at M_1 and M_2 the beams recombine and interfere. Moving M_1 by d changes the path by $2d$; each fringe that crosses the view means $2d = \lambda$ more.

Construction and working

- **S**: broad source (sodium lamp with a ground glass).
- **G_1**: glass plate, half-silvered on the back, at 45°. It splits each wave into two beams of equal amplitude (division of amplitude): one reflected towards M_1, one transmitted towards M_2.
- **M_1, M_2**: plane mirrors at right angles. M_1 moves along its arm on a screw with a fine scale; M_2 is fixed, with tilt screws.
- **G_2**: compensating plate, same thickness and glass as G_1, parallel to it.
- The beams return, recombine at G_1 and go to the telescope T, where they interfere.

> **THE KEY PICTURE**
>
> Seen from the telescope, M_2 appears (by reflection in G_1) as a virtual mirror M_2' beside M_1. The interferometer behaves like an **air film** between M_1 and M_2', of thickness $d = x_1 - x_2$ (the difference of the two arm lengths). Moving M_1 changes d at will.

Circular fringes

1. With M_1 parallel to M_2', it is a parallel air film: path difference $2d\cos\theta$, θ = angle of the ray to the mirror normal
   Cosine law (T13) with n = 1.

2. One beam reflects off the silvered surface of G_1 from the air side, the other from the glass side: they differ by an extra π (Ghatak's convention)
   Stokes, T11.

3. Dark: $2d\cos\theta = m\lambda$; bright: $2d\cos\theta = (m + \frac{1}{2})\lambda$. Equal θ $\Longrightarrow$ rings, Haidinger type (T14), seen focused at infinity

RESULT · MICHELSON

$$2d\cos\theta = m\lambda\ (\text{dark}), \quad 2d\cos\theta = (m + \tfrac{1}{2})\lambda\ (\text{bright})$$

d = separation of M_1 and M_2' (= $x_1 - x_2$), θ = angle of the ray, m = order. Many books ignore the π and write $2d\cos\theta = m\lambda$ for **bright**; say which you use.

- As d **decreases**, each ring shrinks and disappears into the centre; the rings get wider. As d increases, rings come out of the centre.
- Rings near the centre: $\theta_p^2 \approx p\lambda/d$, so radius ∝ √p.

Measuring λ

1. Centre (θ = 0): $2d = m\lambda$ for a dark centre
2. Move M_1 by d_0 and count N fringes passing the centre: $2(d - d_0) = (m - N)\lambda$
3. Subtract: $2d_0 = N\lambda \Longrightarrow \lambda = \dfrac{2d_0}{N}$

   Each fringe = a mirror movement of λ/2 (the light goes there and back).
4. Example: 1000 fringes for $d_0 = 2.90 \times 10^{-2}$ cm: $\lambda = 5.8 \times 10^{-5}$ cm = 5800 Å

Compensating plate

THE IDEA

The beam to M_1 crosses the glass of G_1 three times; without G_2, the beam to M_2 crosses it once. So one beam has an extra glass path $2(n-1)t$. For one wavelength, moving M_1 could make up for it. But n depends on λ, so for **white light** no single mirror position equalises all colours. G_2 gives the other beam the same glass, so the paths are equal for every λ. It is essential for white-light fringes.

Types of fringes

TYPE	MIRRORS	LOOKS LIKE
Circular	$M_1 \parallel M_2'$ (parallel air film), extended source	concentric rings, at infinity (equal inclination)
Straight	M_1 slightly tilted to M_2': an air wedge, d small	straight parallel lines, localised near the mirrors (equal thickness)
Curved	tilted, d larger	arcs (curving away from the edge of the wedge)
White-light	wedge with $d \approx 0$, white source	a few coloured straight fringes round a dark (achromatic) central fringe

Straight and white-light types are general-knowledge material (Ghatak gives circular only); they are standard in Indian textbooks.

WHITE LIGHT FINDS D = 0

White-light fringes appear only when the two paths are equal within a few wavelengths. So they mark exactly the position d = 0 (used in the metre experiment and in measuring $\Delta\lambda$).

Standardising the metre

THE IDEA

Michelson and Benoit (1892–93) measured the old standard metre bar in wavelengths of the red cadmium line, so that the metre could be rebuilt anywhere from light.

1. Nine **etalons** (two mirrors fixed a set distance apart) were made, each half the length of the previous: from 10 cm down to about 0.39 mm.
2. The number of Cd red wavelengths in the smallest etalon was found by counting fringes as M_1 moved from one of its mirrors to the other (white-light fringes mark each mirror position).
3. Each etalon was then compared with one twice its length, by stepping it along; the 10 cm etalon was stepped 10 times along the metre bar.

RESULT

1 m = 1 553 164.13 wavelengths of the red cadmium line (λ = 6438.4696 Å), accurate to about 1 part in 10^9. (Today the metre is defined through the speed of light.)

Etalon details are from Book 2 / general knowledge (Ghatak gives the result only).

In the model paper: **Q23** **Q24** **Q42**.

T21 Michelson: the difference of two wavelengths

> Separating the sodium doublet (PYQ N22 Q20).

1. With white light, set d = 0. Then use sodium light ($\lambda_1 > \lambda_2$, very close). Each makes its own rings; at d = 0 they coincide: sharp fringes

2. Move M_1. The fringes fade when the dark rings of one fall on the bright rings of the other: $\frac{2d}{\lambda_2} - \frac{2d}{\lambda_1} = \frac{1}{2}$

   Same idea as T19.

3. They become sharp again, then fade again when the difference of orders grows by 1. Between two successive fadings M_1 moves Δd with $\frac{2\Delta d}{\lambda_2} - \frac{2\Delta d}{\lambda_1} = 1$

4. $2\Delta d\,\frac{\lambda_1 - \lambda_2}{\lambda_1 \lambda_2} = 1 \Rightarrow \Delta\lambda = \frac{\lambda_1 \lambda_2}{2\Delta d} \approx \frac{\lambda^2}{2\Delta d}$

RESULT

$$\Delta\lambda \approx \frac{\lambda^2}{2\,\Delta d}$$

Δd = mirror movement between two successive disappearances.

1. Example (Ghatak 15.4): Δd = 0.289 mm, λ = 5893 Å
2. $\Delta\lambda = \frac{(5.893\times10^{-7})^2}{2\times2.89\times10^{-4}} = 6.0\times10^{-10}$ m = 6 Å ✓ (5896 − 5890)

EXAM TRAP

Use the distance between two **successive** disappearances (difference of orders = 1). The first disappearance from d = 0 is only half that (difference ½).

LINK TO COHERENCE LENGTH

If the light has a continuous spread Δλ, the fringes vanish for good once $2d \gtrsim \lambda^2/\Delta\lambda$: the coherence length (T2).

In the model paper: **Q38** .

Formula sheet

Phase	$\delta = \dfrac{2\pi}{\lambda}\Delta$
Two waves	$a^2 = a_1^2 + a_2^2 + 2a_1a_2\cos\delta$
Intensity	$I_1 + I_2 + 2\sqrt{I_1 I_2}\cos\delta$; equal: $4I_0\cos^2\dfrac{\delta}{2}$
Visibility	$\dfrac{2\sqrt{I_1 I_2}}{I_1 + I_2}$
Coherence	$L = c\tau \approx \lambda^2/\Delta\lambda$
Young	$\Delta = xd/D$, $\beta = \lambda D/d$
Plane waves	$\beta = \lambda/(2\sin\theta)$
Plate shift	$D(n-1)t/d$
Two mirrors	$d = 2b\theta$, $D = b + c$
Biprism	$d = 2a(n-1)\alpha$, $d = \sqrt{d_1 d_2}$
Lloyd	$d = 2h$; centre dark
Stokes	$r' = -r$, $tt' = 1 - r^2$
Film (air)	refl. dark $2nd\cos\theta' = m\lambda$
AR film	$d = \lambda/4n_f$, $n_f = \sqrt{n_a n_g}$
Wedge	$\beta = \lambda/2n\phi$
Newton	$r_m^2 = m\lambda R/n$; $\lambda = \dfrac{D_{m+p}^2 - D_m^2}{4pR}$
Two λ (rings)	$t_0 = \dfrac{\lambda_1\lambda_2}{4\Delta\lambda}$
Michelson	$2d\cos\theta = m\lambda$; $\lambda = 2d_0/N$
Doublet	$\Delta\lambda = \lambda^2/2\Delta d$

Exam question → topics to know

PYQ	QUESTION (SHORT)	KNOW
N19 Q15, N24 Q4, N25 Q2	Superposition; interference	T1
N22 Q19	Two sinusoids: conditions	T1 M1
N24 Q5, N22 Q13, N20 Q13	Coherence; sustained; division	T2
N19 Q32, N24 Q20, N25 Q3	Intensity; incoherent	T3 T4
N25 Q14, N22 Q7	Fringe width; screen distance	T3
N20 Q19, N19 Q26	Biprism	T8
N22 Q26	Lloyd's mirror	T10
N19 Q5, N19 Q13, N22 Q27	Thin film, normal incidence	T12
N24 Q13, Q14, N25 Q20	Plane film; cosine law	T13
N24 Q16, N20 Q20	Point source; Haidinger	T14
N19 Q22, N24 Q15, N22 Q4	Non-reflecting films	T15
N19 Q14, Q17, N22 Q8, Q14, N24 Q7	Colours; thick plates	T16
N20 Q10, Q31	Wedge	T17
N19 Q3, Q27, N20 Q3, Q26, N22 Q28, Q32	Newton's rings	T18
N19 Q21, N20 Q32, N25 Q4	Michelson; metre; fringes	T20
N22 Q20	Michelson $\Delta\lambda$	T21

Never asked so far, but in the syllabus: two mirrors (T7), white light (T9), Stokes (T11), fringe shift (T6), other geometries (T5), two-λ rings (T19). The model paper covers them.